UNIVERSITY OF AMSTERDAM

REPORT

Assignment 3: Computational Finance

Student:
Julia Rotko, Matteo Postiferi
Student ID: 13603655, 16390288

Lecturer: Sven Karbach

Course:
Computational Finance

March 29, 2026

Contents

1 Introduction **2**

2 Theory **2**
2.1 The Volatility Smile and its Implications for Hedging . . . 2
2.2 The SABR Stochastic Volatility Model . . . 2
2.3 The Hagan Implied Volatility Approximation . . . 3
2.4 Smile-Aware Delta Hedging . . . 3
2.5 Hedging Residuals and Performance Metrics . . . 4

3 Numerical Methods **4**
3.1 Data and Preprocessing . . . 4
3.2 Baseline Black–Scholes Delta . . . 5
3.3 SABR Calibration . . . 5
3.4 SABR and Bartlett Delta Computation . . . 5
3.5 Standardised Filters and Bucketing . . . 5

4 Results and Discussion **6**
4.1 Part 1: Theoretical Derivations . . . 6
4.1.1 Binary Options . . . 6
4.1.2 Knock-out Barrier Options . . . 10
4.2 Part 2: SABR Delta Hedging with Real SPX Data . . . 11
4.2.1 Data Overview and Preprocessing . . . 11
4.2.2 SPX Underlying Series . . . 12
4.2.3 Baseline Black–Scholes Delta Hedging . . . 12
4.2.4 SABR Calibration . . . 12
4.2.5 Delta Comparison and Smile-Aware Hedging . . . 13
4.2.6 Hedging Performance . . . 13

5 Conclusions **14**

6 Figures **15**

1 Introduction

In the first part, we take a theoretical and computational look at the Binary Options and Knock-out Options, and specifically the up-and-out option. In the first part of this section, we derive the theoretical formulas for the Black-Scholes PDE satisfied by the pricing function of the binary option and show the solution to this PDE. Next, we implement computationally the Monte Carlo approach for the binary option. Lastly, we derive the implicit discretization schemes (implicit scheme and Crank-Nicolson scheme) and implement them to compute the option price under these schemes. We also conduct sensitivity analysis, report the option price surface and the delta. In the second part of this subsection, we also take a look at the theoretical price derivation and subsequently computational implementation using Monte Carlo and C-N scheme for the up-and-out option. We comment on the Monte Carlo convergence rates and also report the sensitivity analysis, option price surface, and the delta.

In the second part we evaluate one-step delta hedging strategies for SP 500 (SPX) European call options, extending the analysis beyond the traditional constant-volatility Black-Scholes benchmark. Given that real equity markets exhibit dynamic volatility smiles and pronounced leverage effects, a standard delta hedge leaves a portfolio exposed to significant residual risk. To address this, we implement the lognormal SABR stochastic volatility model and compute Bartlett's delta, which represents the variance-optimal hedge within this framework. By calibrating the SABR model to empirical market data, we compare the hedging performance of the baseline Black-Scholes delta, the smile-aware SABR delta, and the skew-adjusted Bartlett delta using the Sum of Squared Errors (SSE) and a standardized Gain metric.

2 Theory

2.1 The Volatility Smile and its Implications for Hedging

Under the Black–Scholes (BS) model, the implied volatility σ_{imp} extracted from observed option prices should be constant across strikes K and maturities T. In practice, this assumption is systematically violated: implied volatilities exhibit a *skew* — particularly in equity index markets, where out-of-the-money put options command higher implied volatilities than at-the-money options. This pattern, known as the *volatility smile*, reflects both the market's expectation of fat-tailed return distributions and the so-called *leverage effect*: when the underlying index declines, volatility tends to increase, inducing a negative correlation between price and volatility innovations.

The consequence for delta hedging is material. The standard BS delta Δ^{BS} is derived under the assumption of constant volatility, and therefore fails to account for the systematic co-movement between the underlying S_t and the implied volatility surface $\sigma_{\text{imp}}(K, T)$. When implied volatility evolves dynamically, the variance-optimal hedge is no longer the plain delta, but requires a correction proportional to the covariation between the volatility surface and the underlying.

2.2 The SABR Stochastic Volatility Model

To model the joint dynamics of the forward price and its volatility, we adopt the lognormal SABR model [Hagan et al., 2002] with $\beta = 1$ and shift $\theta = 0$. Under the risk-neutral measure, the forward price $F(u)$ and instantaneous volatility $\sigma(u)$ evolve as:

$$dF(u) = \sigma(u)\, F(u)\, dW_1(u), \quad F(0) = F_t, \tag{1}$$

$$d\sigma(u) = \nu\, \sigma(u)\, dW_2(u), \quad \sigma(0) = \alpha, \tag{2}$$

$$d\langle W_1, W_2\rangle_u = \rho\, du, \tag{3}$$

where $\alpha > 0$ is the initial volatility level, $\nu > 0$ is the volatility-of-volatility, and $\rho \in [-1, 1]$ is the instantaneous correlation between forward and volatility shocks. In equity index markets, one expects $\rho < 0$, consistent with the leverage effect.

2.3 The Hagan Implied Volatility Approximation

A closed-form approximation for the SABR implied Black volatility was derived by Hagan et al. [2002]. For the lognormal case $\beta = 1$, defining the log-moneyness $x := \log(F/K)$ and the auxiliary quantities:

$$z := \frac{\nu}{\alpha}\, x, \qquad \xi(z) := \sqrt{1 - 2\rho z + z^2}, \qquad \chi(z) := \log\left(\frac{\xi(z) + z - \rho}{1 - \rho}\right), \tag{4}$$

the implied volatility approximation is:

$$\sigma_{\text{imp}}^{\text{SABR}}(K;\, F, \tau;\, \alpha, \rho, \nu) = \alpha\, \frac{z}{\chi(z)} \left(1 + \left[\frac{\rho\nu\alpha}{4} + \frac{(2 - 3\rho^2)\nu^2}{24}\right]\tau\right), \quad F \neq K, \tag{5}$$

with the stable ATM limit ($F = K$):

$$\sigma_{\text{imp}}^{\text{SABR}}(F;\, F, \tau;\, \alpha, \rho, \nu) = \alpha \left(1 + \left[\frac{\rho\nu\alpha}{4} + \frac{(2 - 3\rho^2)\nu^2}{24}\right]\tau\right). \tag{6}$$

The parameters (α, ρ, ν) are calibrated daily for each maturity slice by minimising the sum of squared errors between market and model implied volatilities.

2.4 Smile-Aware Delta Hedging

Given the SABR implied volatility $\Sigma(F, K, \tau) := \sigma_{\text{imp}}^{\text{SABR}}(K; F, \tau; \hat{\alpha}, \hat{\rho}, \hat{\nu})$, we work with the Black call price on the forward:

$$B(F, K, \tau, \Sigma) = e^{-r\tau}\big(F\,\Phi(d_1) - K\,\Phi(d_2)\big), \quad d_1 = \frac{\ln(F/K) + \frac{1}{2}\Sigma^2\tau}{\Sigma\sqrt{\tau}}, \quad d_2 = d_1 - \Sigma\sqrt{\tau}. \tag{7}$$

We consider three spot-based hedge ratios, each obtained by applying the chain rule $\partial C/\partial S_t = e^{(r-q)\tau}\, \partial B/\partial F$:

Black–Scholes delta. The baseline hedge uses the market implied volatility σ_{mkt} at each strike, ignoring the dynamic dependence of σ_{imp} on F:

$$\Delta_t^{\text{BS}} := e^{(r-q)\tau}\, \frac{\partial B}{\partial F} = e^{-q\tau}\, \Phi(d_1). \tag{8}$$

SABR delta. The smile-aware SABR delta accounts for the sensitivity of the implied volatility to movements in the forward, via the chain rule:

$$\Delta_t^{\text{SABR}} := e^{(r-q)\tau} \left[\frac{\partial B}{\partial F} + \frac{\partial B}{\partial \Sigma}\, \frac{\partial \Sigma}{\partial F}\right], \tag{9}$$

where $\partial B/\partial \Sigma = e^{-r\tau} F \sqrt{\tau}\, \phi(d_1)$ is the Black vega.

Bartlett delta. Bartlett's delta augments the SABR delta with an additional skew correction that accounts for the covariation between the stochastic volatility parameter α and the forward:

$$\Delta_t^{\text{Bart}} := e^{(r-q)\tau}\left[\frac{\partial B}{\partial F} + \frac{\partial B}{\partial \Sigma}\left(\frac{\partial \Sigma}{\partial F} + \frac{\partial \Sigma}{\partial \alpha}\frac{\rho\nu}{F}\right)\right]. \tag{10}$$

As shown in Keller-Ressel et al. [2022], within the lognormal SABR model this hedge coincides with the variance-optimal hedge, i.e. it minimises the mean-square hedging error when implied volatility evolves dynamically. The additional term $(\partial\Sigma/\partial\alpha)\,(\rho\nu/F)$ captures precisely the covariation between the stochastic volatility surface and the underlying — and is proportional to ρ, confirming that its advantage over Δ^{SABR} is most pronounced when the leverage correlation is large in absolute value.

2.5 Hedging Residuals and Performance Metrics

Given a hedge delta δ_t, the one-step hedging residual is defined as:

$$\varepsilon_t(\delta) := \Delta V_t - \delta_t\,\Delta S_t^{\text{hedge}}, \tag{11}$$

where $\Delta V_t = V_{t+1} - V_t$ is the one-day change in the option midquote and $\Delta S_t^{\text{hedge}} = \Delta S_t + S_t(e^{q\Delta t} - 1)$ is the total-return hedge move. The aggregate performance is measured by the Sum of Squared Errors:

$$\text{SSE}(\delta) := \sum_t \varepsilon_t(\delta)^2, \tag{12}$$

and the *Gain* of hedge A relative to benchmark B is defined as:

$$\text{Gain}(A \text{ vs } B) := 1 - \frac{\text{SSE}(\delta^{(A)})}{\text{SSE}(\delta^{(B)})}, \tag{13}$$

where a positive Gain indicates that hedge A reduces the mean-square hedging error relative to B. The competition score is $\text{Gain}(\Delta^{\text{Bart}} \text{ vs } \Delta^{\text{BS}})$, computed after applying the standardised filters described in Section 3.

3 Numerical Methods

3.1 Data and Preprocessing

The dataset consists of SPX European call option quotes sourced from the Wharton Research Data Services (WRDS) database, covering the period 2023-02-01 to 2023-02-28. The raw panel contains 175,301 observations, of which all are call options (`cp_flag = 'C'`). Following standard preprocessing, strikes are rescaled to index points (divided by 1,000), and the option midquote is constructed as $V_t := (\text{bid}_t + \text{ask}_t)/2$. Observations with missing or non-positive implied volatility are removed, yielding 153,728 clean observations across 19 trading dates and 68 distinct expiries.

One-day option changes $\Delta V_t = V_{t+1} - V_t$ are computed within each contract (identified by its ticker symbol), retaining only observations where ΔV_t is defined. This yields 141,619 observations across 11,593 contracts. The SPX closing price series $\{S_t\}$ is obtained from Stooq, covering 2023-02-01 to 2023-03-02 to ensure that one-day changes ΔS_t are available for the last trading day of the sample. The merge between the option panel and the SPX series achieves 100% coverage.

For interest rates and dividends, we adopt flat conventions $r = 4.5\%$ and $q = 1.5\%$ throughout, applied consistently to the forward proxy $F_t = S_t e^{(r-q)\tau_t}$, the BS delta computation, the SABR calibration, and the hedging residuals. The one-day hedge move is computed as the total return:

$$\Delta S_t^{\text{hedge}} := \Delta S_t + S_t\left(e^{q\Delta t} - 1\right), \quad \Delta t = \frac{1}{252}. \tag{14}$$

3.2 Baseline Black–Scholes Delta

The baseline hedge ratio is the spot BS delta computed using the market implied volatility $\sigma_{\mathrm{mkt},t}$ for each individual option:

$$\Delta_t^{\mathrm{BS}} = e^{-q\tau}\,\Phi(d_1), \quad d_1 = \frac{\ln(F_t/K) + \frac{1}{2}\sigma_{\mathrm{mkt},t}^2\,\tau}{\sigma_{\mathrm{mkt},t}\sqrt{\tau}}. \tag{15}$$

This delta serves as the benchmark in all Gain computations. The correlation between our computed Δ^{BS} and the vendor delta provided in the WRDS dataset is 0.9999, confirming the correctness of the implementation.

3.3 SABR Calibration

We adopt the lognormal SABR specification with $\beta = 1$ and shift $\theta = 0$. The choice $\theta = 0$ is appropriate for SPX options, since the underlying index operates in a strictly positive, high-price regime. A non-zero shift is typically introduced in interest rate markets to handle negative rates, which is not a concern here.

For each trading date t and maturity slice $D \in [1, 200]$ days, the SABR parameters $(\hat{\alpha}_{t,D}, \hat{\rho}_{t,D}, \hat{\nu}_{t,D})$ are estimated by nonlinear least squares:

$$(\hat{\alpha}, \hat{\rho}, \hat{\nu}) \in \underset{\alpha>0,\,\nu>0,\,\rho\in[-1,1]}{\arg\min} \sum_{K\in\mathcal{K}_{t,D}} \left(\sigma_{\mathrm{mkt}}(K; t, D) - \sigma_{\mathrm{imp}}^{\mathrm{SABR}}(K; F_t, \tau; \alpha, \rho, \nu)\right)^2, \tag{16}$$

using the full set of available strikes $\mathcal{K}_{t,D}$ for that slice, as required by the assignment specification. The optimisation is performed using the L-BFGS-B algorithm with analytical gradients $\partial\sigma_{\mathrm{imp}}^{\mathrm{SABR}}/\partial\alpha$, $\partial\sigma_{\mathrm{imp}}^{\mathrm{SABR}}/\partial\rho$, $\partial\sigma_{\mathrm{imp}}^{\mathrm{SABR}}/\partial\nu$ derived from Appendix B of the assignment. To mitigate sensitivity to initialisation, three restarts are performed with different starting values of (ρ_0, ν_0), and the solution with the lowest objective value is retained. Parameter bounds are set to $\alpha \in (10^{-4}, 5)$, $\rho \in (-0.999, 0.999)$, $\nu \in (10^{-4}, 5)$. In total, 609 slices are calibrated, all successfully.

3.4 SABR and Bartlett Delta Computation

For each observation, the calibrated parameters $(\hat{\alpha}_{t,D}, \hat{\rho}_{t,D}, \hat{\nu}_{t,D})$ are matched to the option panel via the (t, exdate) key. The partial derivatives $\partial\Sigma/\partial F$ and $\partial\Sigma/\partial\alpha$ required for Δ^{SABR} and Δ^{Bart} are computed using the analytical formulas derived in Appendix B of the assignment, specialised to the lognormal case $\beta = 1$, $\theta = 0$. Specifically, from the simplifications in Section A.10 of the appendix:

$$\Sigma_f := \frac{\partial\Sigma}{\partial F} = (1 + \tau I_H^1)\,\frac{\partial I_B^0}{\partial f}, \qquad \Sigma_\alpha := \frac{\partial\Sigma}{\partial\alpha} = (1 + \tau I_H^1)\,\frac{\partial I_B^0}{\partial\alpha} + \tau I_B^0\,\frac{\rho\nu}{4}, \tag{17}$$

where $I_H^1 = \rho\nu\alpha/4 + (2 - 3\rho^2)\nu^2/24$ is independent of F and K for $\beta = 1$, and $\partial I_B^0/\partial f$, $\partial I_B^0/\partial\alpha$ are the closed-form expressions given in equations (65) and (69) of the appendix. The spot-based hedge ratios are then obtained by multiplying the forward-based expressions by $e^{(r-q)\tau}$.

Note that for the illustrative β-comparison plot in Section 4, where $\beta \in \{0, 0.5, 1\}$, the derivatives $\partial\Sigma/\partial F$ and $\partial\Sigma/\partial\alpha$ for $\beta \neq 1$ are approximated via central differences with step sizes $h_F = 10^{-4}F$ and $h_\alpha = 10^{-4}\hat{\alpha}$, since the analytical implementation in this work is specialised to $\beta = 1$.

3.5 Standardised Filters and Bucketing

Following the competition protocol, observations are filtered after delta computation by removing:

- options with $\Delta_t^{\text{BS}} \leq 0.05$ (deep out-of-the-money),
- options with $\Delta_t^{\text{BS}} \geq 0.95$ (deep in-the-money),
- options with $D_t \leq 14$ days to expiry (near-expiry instability),
- options with $D_t > 200$ days to expiry,
- observations where SABR calibration is unavailable.

The final filtered sample contains 67,285 observations. The filtered panel is partitioned into $9 \times 7 = 63$ buckets using equal-width bins: 9 moneyness bins over $\Delta^{\text{BS}} \in (0.05, 0.95)$ with width 0.10, and 7 maturity bins over $D_t \in (14, 200)$ days with width approximately 26.6 days. SSE, MSE, and Gain metrics are computed both on the full filtered sample and within each bucket.

4 Results and Discussion

4.1 Part 1: Theoretical Derivations

4.1.1 Binary Options

1) Derive the Black-Scholes PDE

By Ito's lemma:

$$dC = \left(\frac{\partial C}{\partial t} + \frac{\partial C}{\partial S} rS + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2}\right) dt + \frac{\partial C}{\partial S}\sigma S \, dB_t$$

Use the change in portfolio:

$$d\Pi = -dV + \Delta \, dS \qquad \Pi = -V + \Delta S$$

Here $V = C$
dS is given:

$$dS = rS \, dt + \sigma S \, dB$$

Plugging in and choosing $\Delta = \frac{\partial C}{\partial S}$:

$$d\Pi = -dC + \frac{\partial C}{\partial S} dS$$

$$d\Pi = -\left[\left(\frac{\partial C}{\partial t} + \frac{\partial C}{\partial S} rS + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2}\right) dt + \frac{\partial C}{\partial S}\sigma S \, dB\right] + \frac{\partial C}{\partial S}(rS \, dt + \sigma S \, dB)$$

$$d\Pi = -\frac{\partial C}{\partial t} dt - \frac{\partial C}{\partial S} rS dt - \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} dt - \frac{\partial C}{\partial S}\sigma S \, dB + \frac{\partial C}{\partial S} rS dt + \frac{\partial C}{\partial S}\sigma S \, dB$$

$$d\Pi = -\frac{\partial C}{\partial t} \, dt - \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} \, dt$$

Risk-neutral condition:

$$d\Pi = r\Pi \, dt = r\left(-C + \frac{\partial C}{\partial S} S\right) dt$$

Equating:

$$-\frac{\partial C}{\partial t} - \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} = -rC + rS\frac{\partial C}{\partial S}$$

$$\Rightarrow \frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC = 0$$

BS PDE:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC = 0$$

Terminal Condition

The terminal condition for the binary (digital) call option is:

$$C(T, S_T) = \mathbf{1}_{\{S_T \geq K\}}$$

Under the risk-neutral measure Q, integrating from 0 to T:

$$\ln(S_T) = \left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}\, z$$

$$\Rightarrow S_T = S_0 \exp\left\{\left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}\, z\right\}$$

Since $S_0 = 1$, this simplifies to:

$$S_T = \exp\left\{\left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}\, z\right\}$$

2) Solution of PDE Proof

Aim: Show that the solution is given by:

$$C_d(t, x) = e^{-r(T-t)}\, \Phi\left(\frac{\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \log\frac{x}{K}}{\sigma\sqrt{T-t}}\right)$$

Digital option price under risk-neutral formula is expressed as:

$$C_d(t, x) = e^{-r(T-t)} \mathbb{E}^Q\left[\mathbf{1}_{\{S_T \geq K\}} \mid S_t = x\right]$$

Using the representation:

$$S_T = x \exp\left\{\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma\sqrt{T-t}\, z\right\}, \quad z \sim N(0,1)$$

This is the price representation to which we arrived before, just instead of having integrated over (0, T), we integrate here over (t, T). Given the variable z we then have:

$$C_d(t, x) = e^{-r(T-t)} \int_{-\infty}^{\infty} \mathbf{1}_{\{S_T(z) \geq K\}} \varphi(z)\, dz$$

Condition:

$$S_T \geq K$$

$$x \exp\left\{\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma\sqrt{T-t}\, z\right\} \geq K$$

Taking logs:

$$\ln x + \left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma\sqrt{T-t}\, z \geq \ln K$$

$$\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma\sqrt{T-t}\,z \geq \ln\left(\frac{K}{x}\right)$$

Rearranging:

$$\sigma\sqrt{T-t}\,z \geq \ln\left(\frac{K}{x}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-t)$$

$$z \geq \frac{\ln\left(\frac{K}{x}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}$$

Define:

$$d_-(T-t) = \frac{\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \ln\left(\frac{x}{K}\right)}{\sigma\sqrt{T-t}}$$

Then:

$$C_d(t,x) = e^{-r(T-t)} \int_{d_-}^{\infty} \varphi(z)\,dz$$

Using the standard normal CDF:

$$\int_{d_-}^{\infty} \varphi(z)\,dz = \Phi(-d_-) = \Phi(d_-^*)$$

with:

$$d_-^* = d_-(T-t)$$

Therefore:

$$C_d(t,x) = e^{-r(T-t)}\Phi(d_-(T-t))$$

3) Monte Carlo

We implemented the computation of the binary option with Monte Carlo, using $N = 10000$ MC runs and computing the final price of the option as the following estimator:

$$\hat{C} = e^{-rT} \cdot \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{\left\{S_T^{(i)} > K\right\}}$$

The resulting Monte Carlo binary option price of our simulation was 0.2087.

4) Implicit Discretization Scheme

We have our PDE:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC = 0$$

The grid:

$$S_i = i\,\Delta S, \qquad t_n = n\,\Delta t, \qquad C_i^n = C(S_i,\, t_n)$$

Derivatives:

$$\frac{\partial C}{\partial t} \approx \frac{C_i^{n+1} - C_i^n}{\Delta t}, \qquad \frac{\partial C}{\partial S} \approx \frac{C_{i+1}^{n+1} - C_{i-1}^{n+1}}{2\Delta S}, \qquad \frac{\partial^2 C}{\partial S^2} \approx \frac{C_{i+1}^{n+1} - 2C_i^{n+1} + C_{i-1}^{n+1}}{\Delta S^2}$$

Plug into the PDE:

$$\frac{C_i^{n+1} - C_i^n}{\Delta t} + \frac{1}{2}\sigma^2 S_i^2 \left(\frac{C_{i+1}^{n+1} - 2C_i^{n+1} + C_{i-1}^{n+1}}{\Delta S^2}\right) + rS_i\left(\frac{C_{i+1}^{n+1} - C_{i-1}^{n+1}}{2\Delta S}\right) - rC_i^{n+1} = 0$$

Multiply by Δt:

$$C_i^{n+1} - C_i^n + \frac{1}{2}\sigma^2 S_i^2 \Delta t \left(\frac{C_{i+1}^{n+1} - 2C_i^{n+1} + C_{i-1}^{n+1}}{\Delta S^2}\right) + rS_i\Delta t\left(\frac{C_{i+1}^{n+1} - C_{i-1}^{n+1}}{2\Delta S}\right) - r\Delta t\, C_i^{n+1} = 0$$

C_{i+1}^{n+1} terms:

$$C_{i+1}^{n+1}\left[\frac{1}{2}\sigma^2 S_i^2 \frac{\Delta t}{\Delta S^2} + rS_i \frac{\Delta t}{2\Delta S}\right] \quad \leftarrow a_i$$

C_i^{n+1} terms:

$$C_i^{n+1}\left[1 - \sigma^2 S_i^2 \frac{\Delta t}{\Delta S^2} - r\Delta t\right] \quad \leftarrow b_i$$

C_{i-1}^{n+1} terms:

$$C_{i-1}^{n+1}\left[\frac{1}{2}\sigma^2 S_i^2 \frac{\Delta t}{\Delta S^2} - rS_i \frac{\Delta t}{2\Delta S}\right] \quad \leftarrow c_i$$

So we have the coefficients:

$$a_i = \Delta t\left[\frac{1}{2}\sigma^2 \frac{S_i^2}{\Delta S^2} + \frac{rS_i}{2\Delta S}\right]$$

$$b_i = 1 - \Delta t\left[\sigma^2 \frac{S_i^2}{\Delta S^2} + r\right]$$

$$c_i = \Delta t\left[\frac{1}{2}\sigma^2 \frac{S_i^2}{\Delta S^2} - \frac{rS_i}{2\Delta S}\right]$$

Final system:

$$a_i C_{i+1}^{n+1} + b_i C_i^{n+1} + c_i C_{i-1}^{n+1} = C_i^n$$

4) Crank-Nicolson Scheme

We use a spatial operator $\mathcal{L}(C)$

The PDE:

$$\frac{\partial C}{\partial t} + \underbrace{\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC}_{\mathcal{L}(C)} = 0$$

So:

$$\frac{\partial C}{\partial t} + \mathcal{L}(C) = 0$$

Crank Nicolson idea is that we average the $n+1$ and n:

$$\mathcal{L} \approx \frac{1}{2}\left(\mathcal{L}^{n+1} + \mathcal{L}^n\right)$$

$$\frac{\partial C}{\partial t} \approx \frac{C_i^{n+1} - C_i^n}{\Delta t}$$

$$\Rightarrow \frac{C_i^{n+1} - C_i^n}{\Delta t} + \frac{1}{2}\left(\mathcal{L}^{n+1}C_i^{n+1} + \mathcal{L}^n C_i^n\right) = 0$$

$$C_i^{n+1} - C_i^n + \frac{\Delta t}{2}\mathcal{L}^{n+1}C_i^{n+1} + \frac{\Delta t}{2}\mathcal{L}^n C_i^n = 0$$

$$\mathcal{L}C_i = a_i C_{i-1} + b_i C_i + c_i C_{i+1}$$

$$\Rightarrow C_i^{n+1} - C_i^n + \frac{\Delta t}{2}\left(a_i C_{i-1}^{n+1} + b_i C_i^{n+1} + c_i C_{i+1}^{n+1}\right) + \frac{\Delta t}{2}\left(a_i C_{i-1}^n + b_i C_i^n + c_i C_{i+1}^n\right) = 0$$

$$\frac{\Delta t}{2} a_i C_{i-1}^{n+1} + \left(1 + \frac{\Delta t}{2} b_i\right) C_i^{n+1} + \frac{\Delta t}{2} c_i C_{i+1}^{n+1} = C_i^n - \frac{\Delta t}{2}\left(a_i C_{i-1}^n + b_i C_i^n + c_i C_{i+1}^n\right)$$

$$\Rightarrow A_i C_{i-1}^{n+1} + B_i C_i^{n+1} + C_i C_{i+1}^{n+1} = D_i$$

where:

$$A_i = \frac{\Delta t}{2} a_i, \quad B_i = 1 + \frac{\Delta t}{2} b_i, \quad C_i = \frac{\Delta t}{2} c_i$$

$$D_i = C_i^n - \frac{\Delta t}{2}\left(a_i C_{i-1}^n + b_i C_i^n + c_i C_{i+1}^n\right)$$

The finite difference schemes were implemented to price the binary option. The price obtained using both the Implicit Scheme and Crank-Nicolson Scheme was 0.4756.

We also report the sensitivity analysis to the parameter K, which can be read in the Figure 1. Figure 2 represents the binary option price with respect to time t and the stock price S_t. Lastly, we compute the delta as a function of the underlying S and report it in the Figure 3.

4.1.2 Knock-out Barrier Options

1) Option Price Derivation

We want to price the up-and-out barrier call option. The payoff is:

$$C = (S_T - K)^+ \mathbf{1}_{\max_{0 \le t \le T} S_t < B}$$

The fair value is the discounted expected payoff under the risk-neutral measure:

$$C(t, T, S_t, M_0^t) := e^{-(T-t)r}\, \mathbb{E}\left[(S_T - K)^+ \mathbf{1}_{\{M_T^t < B\}} \,\middle|\, \mathcal{F}_t\right]$$

We split the expectation using $\mathbb{E}[(S_T - K)^+] = \mathbb{E}[S_T \cdot \mathbf{1}_{S_T > K}] - K\,\mathbb{E}[\mathbf{1}_{S_T > K}]$, and we apply the reflection principle to get the probability of paths that crossed B at some point in the trajectory, but ended the trajectory below B at maturity. These paths must be subtracted because they don't contribute to the payoff as the option expired by triggering the barrier.

$$= S_t\, \mathbf{1}_{\{M_0^t < B\}} \left\{\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{K}\right)\right) - \Phi\left(\delta_+^{T-t}\left(\frac{S_t}{B}\right)\right)\right\}$$

$$- S_t\, \mathbf{1}_{\{M_0^t < B\}} \left(\frac{B}{S_t}\right)^{1+2r/\sigma^2} \left\{\Phi\left(\delta_+^{T-t}\left(\frac{B^2}{K S_t}\right)\right) - \Phi\left(\delta_+^{T-t}\left(\frac{B}{S_t}\right)\right)\right\}$$

$$- e^{-(T-t)r} K\, \mathbf{1}_{\{M_0^t < B\}} \left\{\Phi\left(\delta_-^{T-t}\left(\frac{S_t}{K}\right)\right) - \Phi\left(\delta_-^{T-t}\left(\frac{S_t}{B}\right)\right)\right\}$$

$$+ \left(\frac{S_t}{B}\right)^{1-2r/\sigma^2} e^{-(T-t)r} K\, \mathbf{1}_{\{M_0^t < B\}} \left\{\Phi\left(\delta_-^{T-t}\left(\frac{B^2}{K S_t}\right)\right) - \Phi\left(\delta_-^{T-t}\left(\frac{B}{S_t}\right)\right)\right\}$$

where:

$$\delta_\pm^\tau(z) = \frac{1}{\sigma\sqrt{\tau}}\left(\log z + \left(r \pm \frac{\sigma^2}{2}\right)\tau\right), \quad z > 0$$

The variable z takes different possible values to reflect the given probability: $\frac{S_t}{K}$ means how far the stock price is from the strike, $\frac{S_t}{B}$ - how far the stock price is from the barrier, $\frac{B^2}{K S_t}$ - how far the reflected starting stock price is from the strike, and lastly $\frac{B}{S_t}$ - probability that the stock price reaches exactly B at maturity.

2) Comparison Analytical Formula and Monte Carlo

In this part, we calculate the up-and-out option price with Monte Carlo simulations and compare it to the analytical result. We implement the analytical formula and get a resulting price of 1.1761. For the Monte Carlo, we run the simulations for different values of discrete number of steps $m = [10, 20, 50, 100, 250, 500]$ to study the convergence rates. The reason why looking into convergence rates with MC is important is because the Monte Carlo method tends to underestimate the number of times the barrier was hit. As the method only computes values at m discrete time-steps, the hit to the barrier B can occur between two such time-steps and therefore not be recorded. Thus, we use an adjusted method where we adjust the barrier B and also study the convergence. The prices we get for the different values of m are, respectively: 2.9470, 2.4092, 1.9030, 1.6903, 1.4927, and 1.3742, with corresponding errors of 1.7709, 1.2331, 0.7269, 0.5142, 0.3167, and 0.1981. Thus, as can be seen, with the larger number of steps, the price gets closer to its analytical result. This is consistent with the theoretical convergence rate of $1/\sqrt{m}$, meaning that to halve the error you must quadruple the number of monitoring steps.

We also report the sensitivity to the option and model parameters (S, σ and B) for the analytical approach in the Figure 4.

3) Implicit Discretization Scheme

We report the sensitivity analysis to the option and model parameters using Crank-Nicolson scheme in the Figure 5. Next, we also report the option price surface for the up-and-out option in the Figure 6. Lastly, we report the delta as a function of the stock price S and report the results in the Figure 7.

4.2 Part 2: SABR Delta Hedging with Real SPX Data

4.2.1 Data Overview and Preprocessing

The raw WRDS dataset contains 175,301 call option observations spanning 19 trading dates and 68 distinct expiries over February 2023. After removing observations with missing or non-positive implied volatility, 153,728 observations remain, corresponding to a missing rate of 12.31% for both σ_{mkt} and δ_{vendor}. The midquote V_t is always well-defined, with a missing rate of 0%. The columns retained for the analysis are: `date`, `exdate`, `symbol`, `strike_price`, `best_bid`, `best_offer`, `V`, `impl_volatility`, `delta`.

After constructing one-day option changes ΔV_t within each contract and retaining only observations where ΔV_t is defined, the panel reduces to 141,619 observations across 11,593 contracts. Summary statistics of the midquote and its daily change are reported in Table 1. The mean option value of 429.6 index points reflects the dominance of deep in-the-money options in the raw panel. The mean daily change of -6.32 points is consistent with the broadly declining SPX over the sample period.

Statistic	V_t	ΔV_t
Mean	429.63	−6.32
Std	583.42	27.13
Min	0.03	−170.30
25%	33.90	−22.20
Median	202.35	−2.15
75%	596.35	5.10
Max	4022.70	70.20

Table 1: Summary statistics of option midquote V_t and one-day change ΔV_t (index points), computed on 141,619 observations.

4.2.2 SPX Underlying Series

Figure 8 shows the SPX closing price S_t and its one-day changes ΔS_t over the sample period. The index declined from approximately 4,180 at the start of February to 3,950 by end of month, a cumulative drop of roughly 5.5%. The mean daily change is -6.89 points with a standard deviation of 38.25 points, reflecting a period of elevated but not extreme volatility. The most notable single-day move is -81.75 points on 16 February 2023, coinciding with hotter-than-expected inflation data. The SPX series is merged into the option panel with 100% coverage across all 141,619 observations.

4.2.3 Baseline Black–Scholes Delta Hedging

Figure 9 shows $\Delta^{\mathrm{BS}}(K)$ for the maturity slice of 14 February 2023, expiry 15 March 2023 (29 calendar days). The computed BS delta closely tracks the vendor delta provided in the WRDS dataset, with a Pearson correlation of 0.9999 across all 141,619 observations, confirming the correctness of the implementation. The sigmoid shape is consistent with Black–Scholes theory: $\Delta^{\mathrm{BS}} \approx 1$ for deep in-the-money options ($K \ll S$), $\Delta^{\mathrm{BS}} \approx 0.5$ at-the-money ($K \approx S = 4{,}136$), and $\Delta^{\mathrm{BS}} \approx 0$ for deep out-of-the-money options.

The hedging residuals $\varepsilon_t(\Delta^{\mathrm{BS}})$ have mean 1.03 and standard deviation 39.21 index points on the pre-filter panel, indicating that the hedge is approximately unbiased. The baseline SSE after the standardised Step 6 filters is $\mathrm{SSE}(\Delta^{\mathrm{BS}}) = 99{,}372{,}918$ on 67,285 observations.

Volatility misspecification experiment. Figure 10 reports $\mathrm{SSE}(\sigma)$ as a function of a flat volatility σ applied uniformly to all options, computed on the pre-filter panel. The curve exhibits the expected U-shape with a minimum at $\sigma^* = 0.775$, substantially above the mean market implied volatility of 0.253. This divergence arises because the pre-filter panel contains a large proportion of deep in-the-money options whose delta is insensitive to σ, thereby biasing the optimal flat volatility upward. Crucially, this experiment confirms that using the option-specific σ_{mkt} — as done in Δ^{BS} — is superior to any fixed flat volatility, since it adapts to the cross-sectional variation of the smile.

4.2.4 SABR Calibration

Smile fit. Figure 11 shows the market implied volatility smile and the SABR fit for the slice of 23 February 2023, expiry 17 March 2023 (22 calendar days), with calibrated parameters $\hat{\alpha} = 0.143$, $\hat{\rho} = -0.656$, $\hat{\nu} = 5.000$. The SABR model reproduces the near-ATM smile accurately, where the implied volatility minimum occurs around $K \approx 4{,}200$ and rises steeply for lower strikes. The negative value of $\hat{\rho}$ is consistent with the equity leverage effect: when the index declines, implied volatility increases, generating the observed downward skew. The visible misfit at very low strikes ($K < 1{,}000$) reflects the known limitation of the Hagan approximation far from the money, which is not a concern in practice since these options are excluded by the Step 6 filters.

Parameter time series. Figure 12 shows the time series of $\hat{\alpha}_{t,D}$ and $\hat{\nu}_{t,D}$ for maturities $D = 21$ and $D = 28$ days. The volatility level $\hat{\alpha}$ rises from approximately 0.15 at the start of February to 0.19–0.22 by end of month, mirroring the declining SPX trend and the associated increase in realised volatility. The two maturity slices track each other closely, indicating temporal consistency of the calibration. The vol-of-vol parameter $\hat{\nu}$ is more volatile, oscillating between 2.0 and 5.0, which reflects the sensitivity of the smile curvature to day-to-day market conditions.

Calibration error. Figure 13 shows the distribution of $\mathrm{RMSE}_{\mathrm{vol}}$ across all 609 calibrated slices. The distribution is strongly right-skewed: the median RMSE of 0.013 indicates that

the majority of slices are fitted with high accuracy, while the mean of 0.035 is inflated by a tail of poorly-fitted slices. These high-error slices correspond almost exclusively to short-dated maturities ($D \leq 14$ days), where the Hagan approximation is known to deteriorate and where the implied volatility surface takes a steep V-shape that a smooth parametric model cannot replicate. All such slices are removed by the standardised Step 6 filter $D > 14$.

4.2.5 Delta Comparison and Smile-Aware Hedging

Delta profiles. Figure 14 compares $\Delta^{\mathrm{BS}}(K)$, $\Delta^{\mathrm{SABR}}(K)$, and $\Delta^{\mathrm{Bart}}(K)$ for the slice of 14 February 2023, expiry 15 March 2023 ($D = 29$ days, $F = 4{,}146$, calibrated $\hat{\rho} = -0.530$ for this slice). The three deltas exhibit a clear and economically interpretable ordering:

$$\Delta^{\mathrm{Bart}} < \Delta^{\mathrm{BS}} < \Delta^{\mathrm{SABR}},$$

holding uniformly across all strikes. The SABR delta lies above Δ^{BS} because, when F increases, the implied volatility smile shifts upward via $\partial\Sigma/\partial F > 0$, further increasing the option value and thus the hedge ratio. Bartlett's delta lies below Δ^{BS} because the additional covariation term $(\partial\Sigma/\partial\alpha)(\rho\nu/F)$ is negative for $\hat{\rho} = -0.530 < 0$: when F rises, α tends to fall due to the leverage effect, partially offsetting the direct price sensitivity. The separation at-the-money is significant: Δ^{SABR} is approximately 8 percentage points higher than Δ^{BS}, while Δ^{Bart} is approximately 6 percentage points lower, a difference that is economically material for practical hedging.

Effect of β. Figure 15 shows the effect of $\beta \in \{0, 0.5, 1\}$ on $\Delta^{\mathrm{SABR}}(K)$ and $\Delta^{\mathrm{Bart}}(K)$, computed at the parameters calibrated for $\beta = 1$. For $\beta = 1$ (lognormal SABR), both deltas exhibit smooth sigmoid profiles consistent with standard option theory. For $\beta \in \{0, 0.5\}$, the profiles degenerate into near-step functions, resembling the delta of a digital option rather than a European call. This behaviour arises because the SABR model with $\beta < 1$ implies a distribution for the forward with heavier tails and different absorption properties at zero; when the parameters (α, ρ, ν) are not recalibrated to the new β, the implied volatility surface becomes numerically unstable far from the money, driving d_1 to extreme values and collapsing the delta to a step function. This result underscores the importance of joint calibration of all model parameters including β, and motivates the assignment's choice to fix $\beta = 1$ for a stable and well-identified specification.

4.2.6 Hedging Performance

Aggregate results. Table 2 summarises the total SSE, MSE, and Gain metrics on the final filtered sample of 67,285 observations.

	Δ^{BS}	Δ^{SABR}	Δ^{Bart}
SSE	99,372,918	108,258,717	92,313,646
MSE	1,476.9	1,609.0	1,372.0
Gain vs BS (%)	—	−8.9%	+7.1%

Table 2: Total SSE, MSE, and Gain on the filtered sample ($n = 67{,}285$). Gain is defined as $1 - \mathrm{SSE}(\delta^{(A)})/\mathrm{SSE}(\Delta^{\mathrm{BS}})$.

Bartlett's delta achieves a Gain of +7.1% over the BS benchmark, reducing the aggregate mean-square hedging error from 1,476.9 to 1,372.0. The plain SABR delta, by contrast, performs worse than BS (Gain = −8.9%), which is consistent with the theoretical prediction of Keller-Ressel et al. [2022]: the smile-aware chain-rule correction alone is insufficient to improve hedging

performance without the Bartlett covariation adjustment. Only when the leverage-effect covariation between α and F is explicitly incorporated does the delta become variance-optimal and outperform the BS benchmark.

Bucketed analysis. Figure 16 shows the Gain of Bartlett vs BS (left) and the relative Gain of Bartlett vs SABR (right) across the 9×7 moneyness–maturity grid. Several patterns are evident. First, all 63 buckets exhibit positive Gain for Bartlett vs BS, confirming the robustness of the result across the full option spectrum. Second, the Gain generally increases with maturity: for near-ATM options (moneyness bins 4–5), the Gain rises from approximately 0.17 in the shortest maturity bin to 0.20–0.21 in the longest, though with some noise across adjacent bins. This is consistent with theory — stochastic volatility effects accumulate over time, making the covariation correction more valuable for longer-dated options. Third, the Gain is highest for options closest to at-the-money (moneyness bins 1–3, corresponding to $\Delta^{\mathrm{BS}} \in [0.05, 0.35]$), reaching values up to 0.46 in some buckets. This reflects the fact that at-the-money options have the highest vega, amplifying the impact of the volatility correction on the hedge ratio. The relative Gain of Bartlett vs SABR (right panel) shows an even stronger pattern, with values up to 0.50 in the near-ATM, long-maturity corner, confirming that the Bartlett correction delivers substantial improvement over the plain SABR delta across all market conditions.

5 Conclusions

In the first part of this assignment, we analyzed, both theoretically and computationally, the dynamics of the binary option and the knock-out option (up-and-out option). An important note is specifically about the convergence rates in the Monte Carlo implementation for the up-and-out option price - as the evaluations happen at discrete timesteps m, the barrier can be hit between them and come unnoticed. Therefore, it is important to choose a large enough number of steps such that the price is close to its analytical result.

In the second part we evaluated three delta hedging strategies for SPX European call options over February 2023: the Black–Scholes delta Δ^{BS}, the smile-aware SABR delta Δ^{SABR}, and Bartlett's variance-optimal delta Δ^{Bart}.

The main finding is that Bartlett's delta reduces the aggregate mean-square hedging error by 7.1% relative to the BS benchmark, with positive gains across all 63 moneyness–maturity buckets. The improvement is largest for near-ATM, longer-dated options, where vega is highest and stochastic volatility effects are most material. The plain SABR delta underperforms BS by 8.9%, confirming that the smile-aware correction without the Bartlett covariation adjustment is insufficient — and indeed harmful. These results are consistent with the theoretical prediction of Keller-Ressel et al. [2022] that Bartlett's delta is the variance-optimal hedge within lognormal SABR, with its advantage most pronounced when $|\rho|$ is large.

The main limitations are the use of flat interest rates and dividend yield, a single-month sample period characterised by a declining market, and the absence of transaction costs. Extending the analysis to longer horizons, time-varying rates, and alternative stochastic volatility models represents a natural direction for future work.

6 Figures

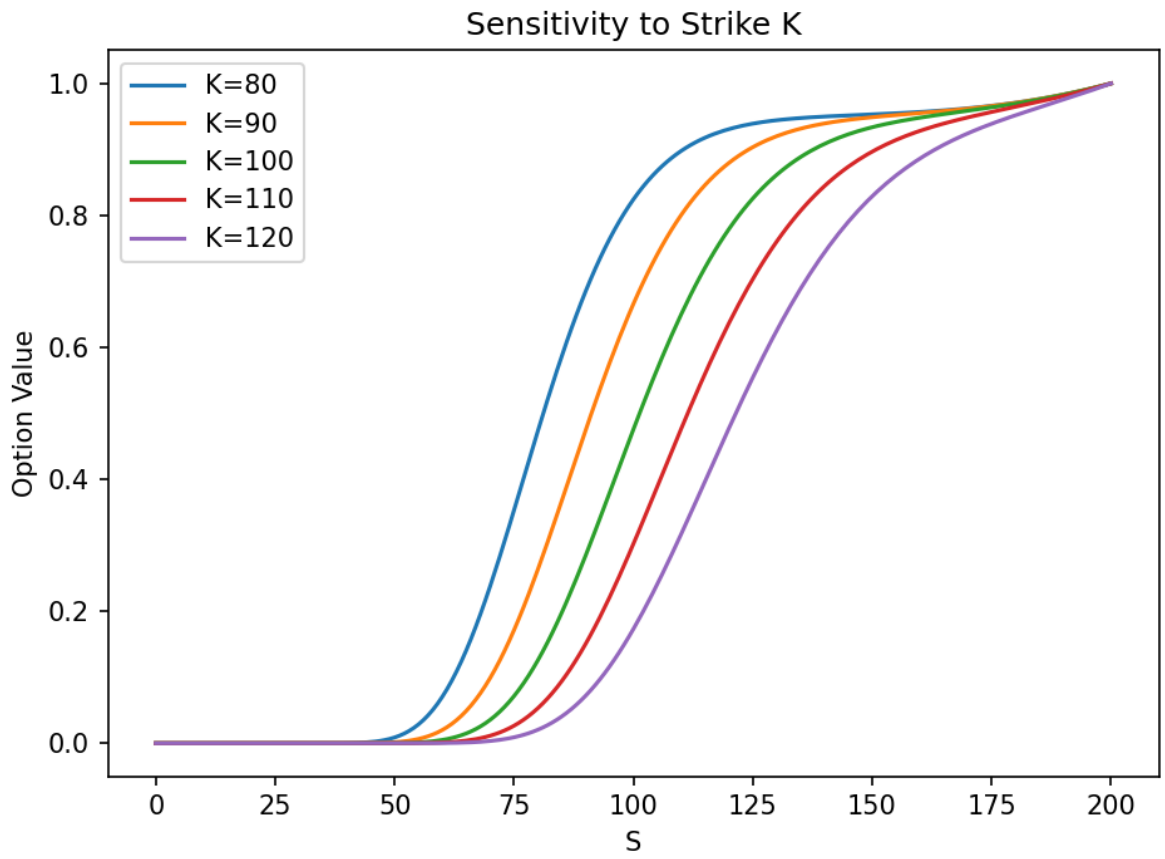

Figure 1: Sensitivity of the binary option price to the strike parameter K, computed using the Implicit Scheme and the Crank–Nicolson scheme.

References

Hagan, P. S., Kumar, D., Lesniewski, A. S., & Woodward, D. E. (2002). Managing smile risk. *Wilmott Magazine*, 1(1), 84-108.

Keller-Ressel, M., & Kilin, F. (2022). Bartlett's delta in the SABR model. *Quantitative Finance*, 22(11), 1989-2005.

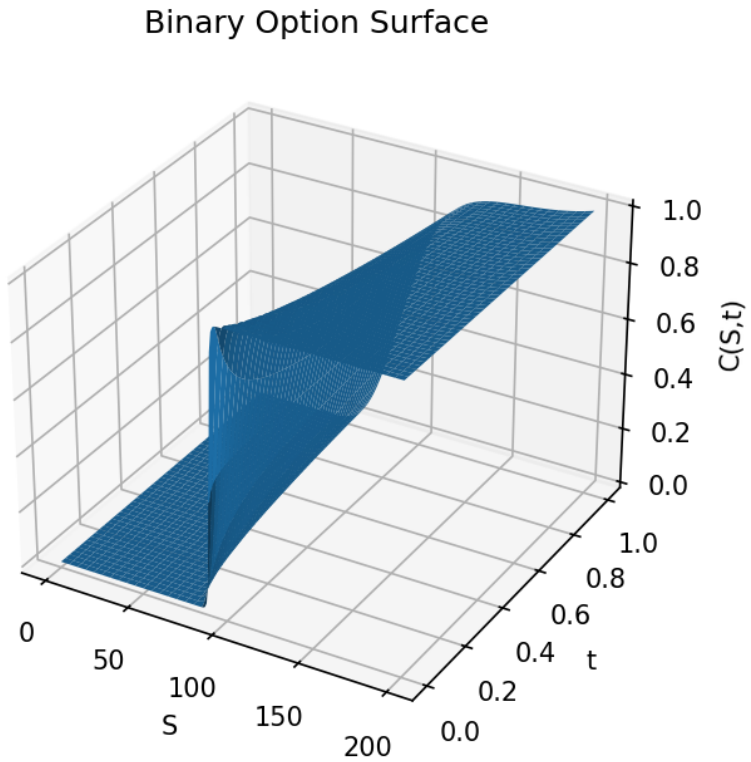

Figure 2: Price surface $C(S,t)$ of the binary option computed using finite difference approximations.

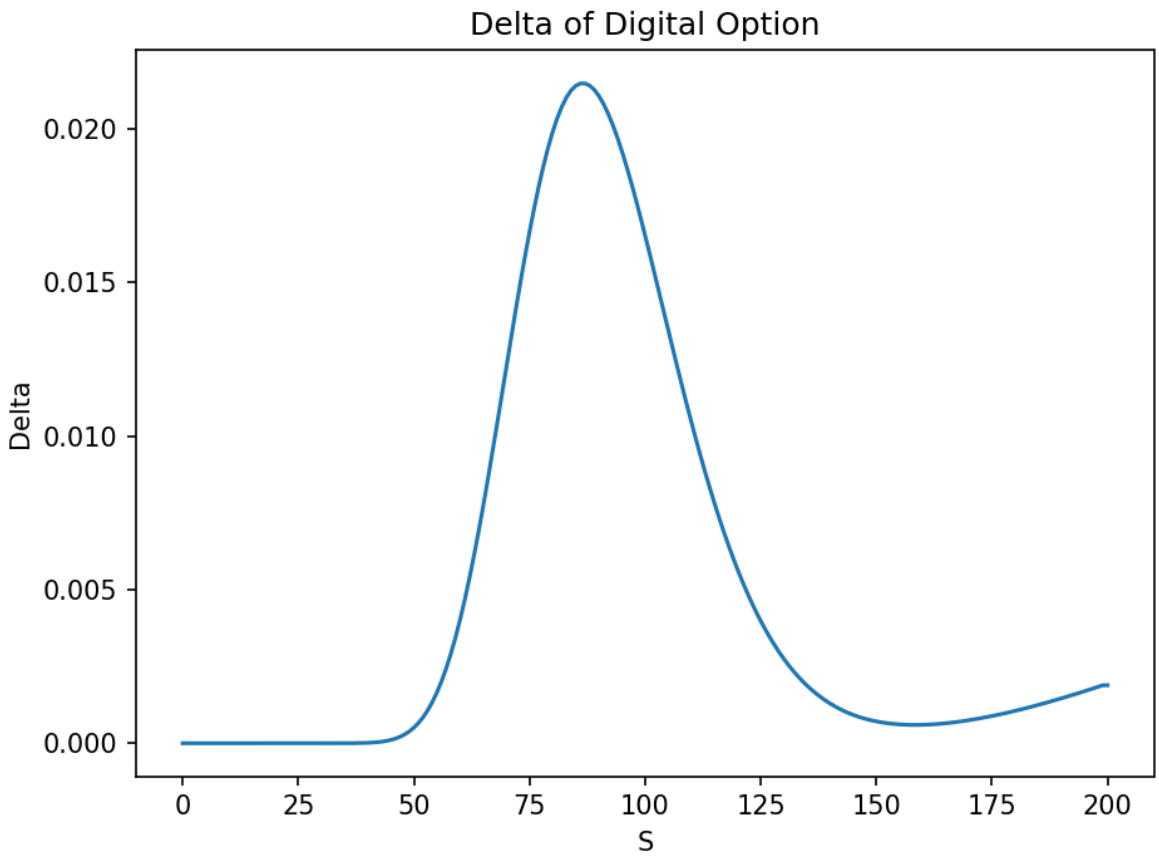

Figure 3: Delta of the binary option as a function of the underlying price S.

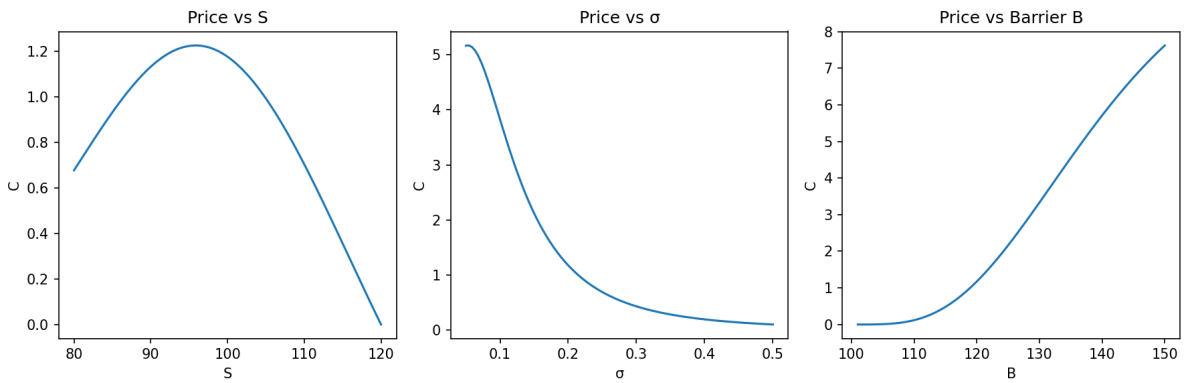

Figure 4: Sensitivity of the up-and-out barrier call option price to model and contract parameters, computed using the analytical formula.

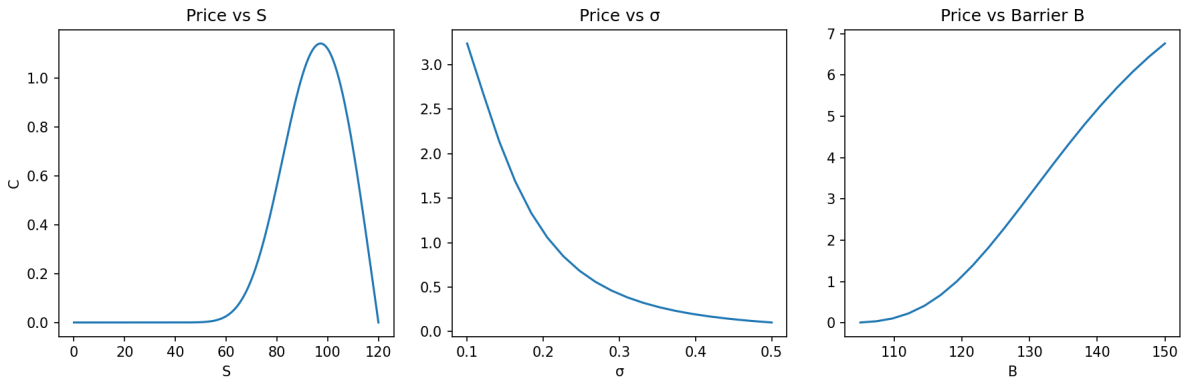

Figure 5: Sensitivity of the up-and-out barrier call option price to model and contract parameters, computed using the Crank–Nicolson scheme.

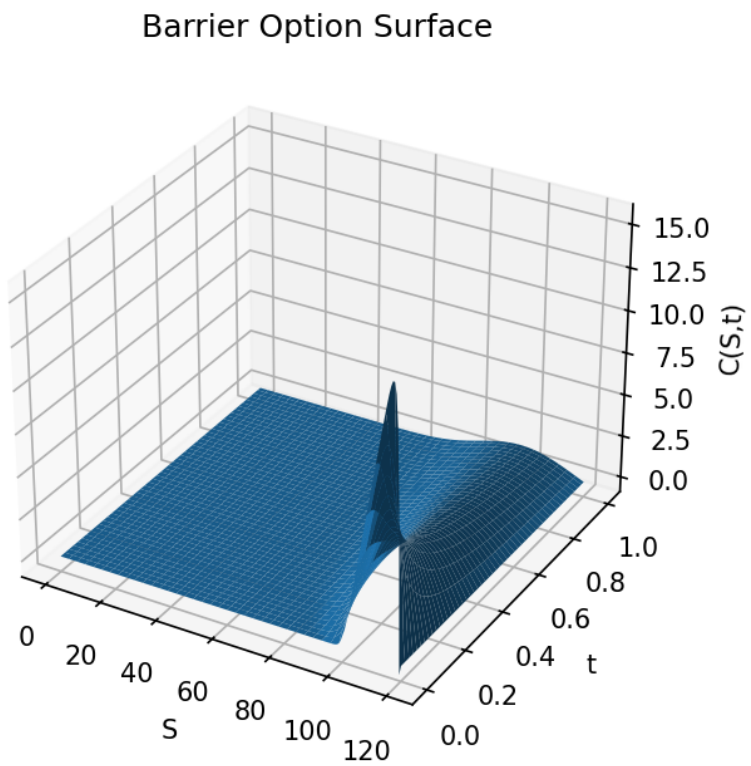

Figure 6: Price surface $C(S,t)$ of the up-and-out barrier call option computed using the Crank–Nicolson scheme.

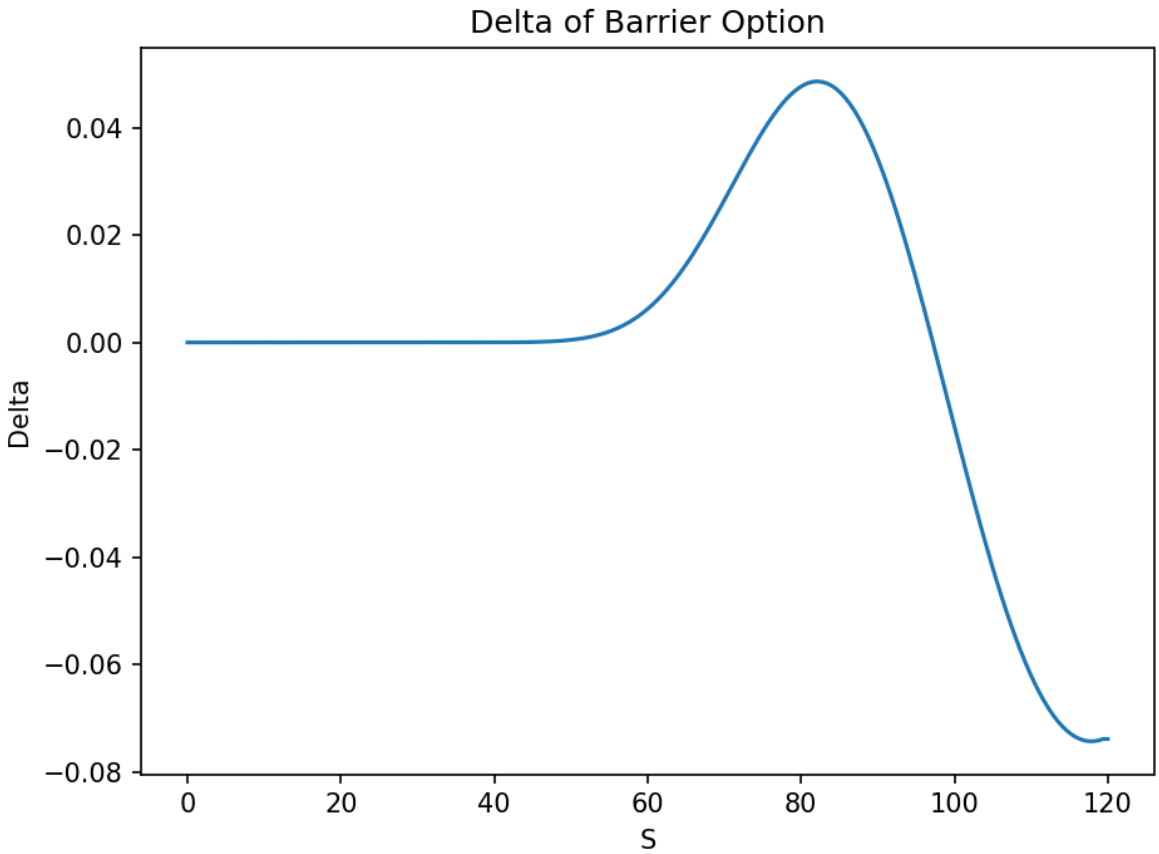

Figure 7: Delta of the up-and-out barrier call option as a function of the underlying price S.

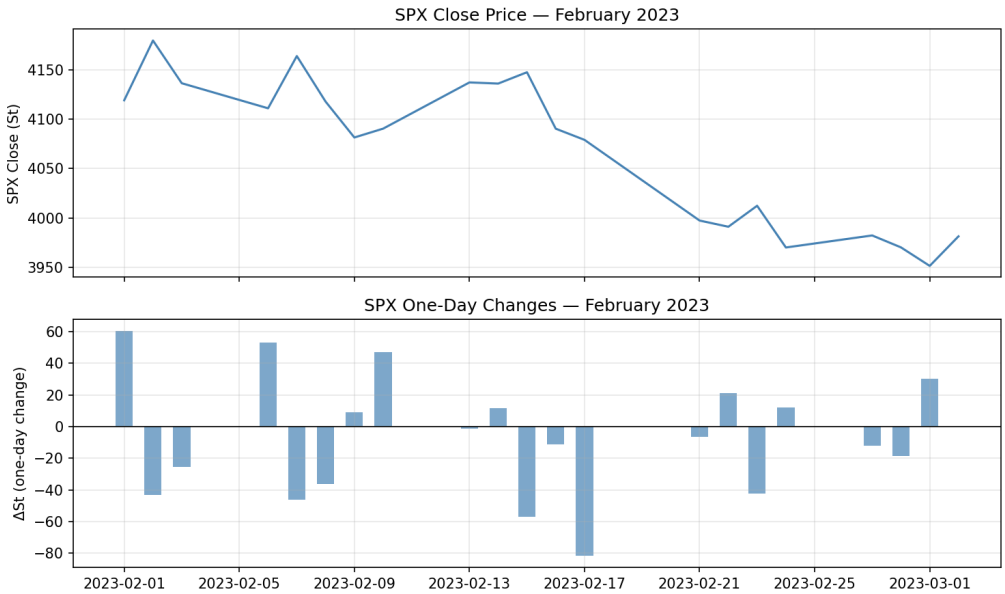

Figure 8: SPX closing price S_t (top) and one-day changes ΔS_t (bottom) over February 2023.

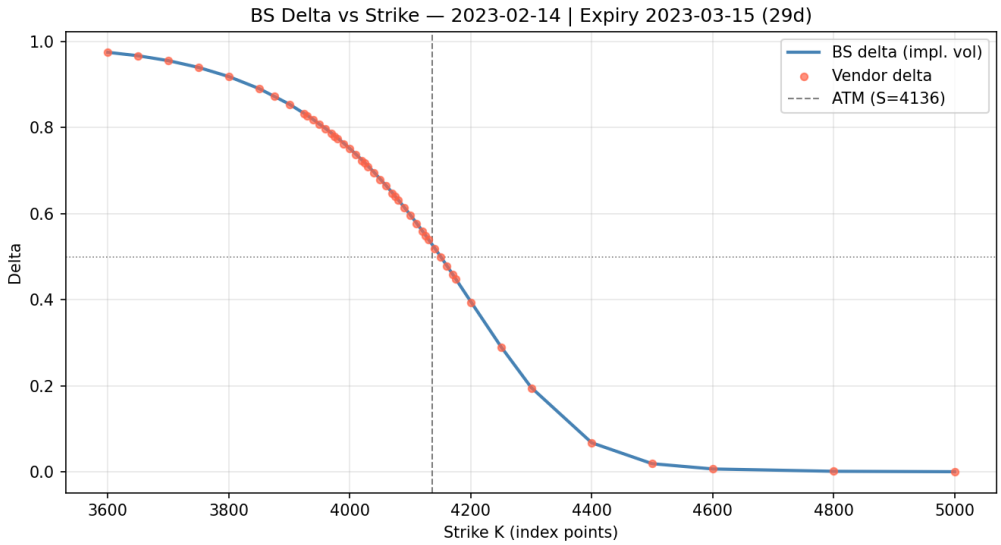

Figure 9: Black–Scholes delta $\Delta^{\mathrm{BS}}(K)$ (blue line) and vendor delta (orange dots) as a function of strike K. Date: 14 February 2023, expiry 15 March 2023 (29 days).

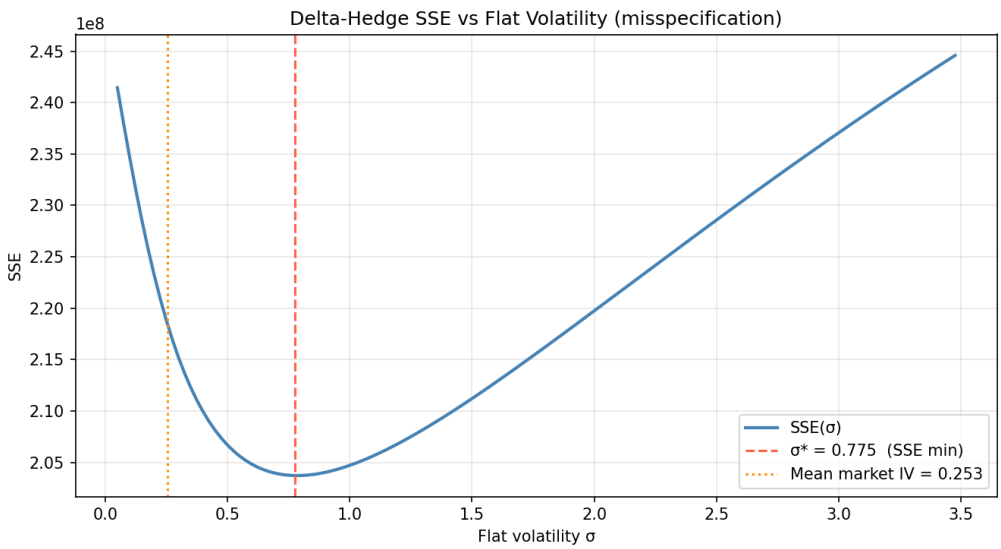

Figure 10: Hedging SSE as a function of a flat volatility σ applied uniformly to all options (pre-filter panel). The dashed red line marks $\sigma^* = 0.775$; the dotted orange line marks the mean market implied volatility of 0.253.

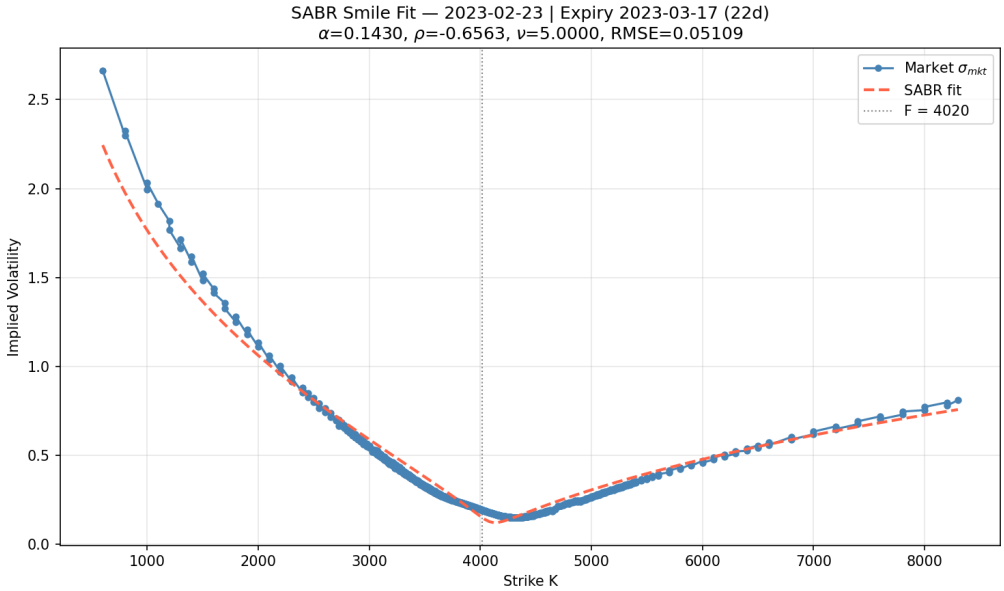

Figure 11: Market implied volatility smile (blue) and SABR fitted smile (red dashed) as a function of strike K. Date: 23 February 2023, expiry 17 March 2023 (22 days). Calibrated parameters: $\hat{\alpha} = 0.143$, $\hat{\rho} = -0.656$, $\hat{\nu} = 5.000$, RMSE $= 0.051$.

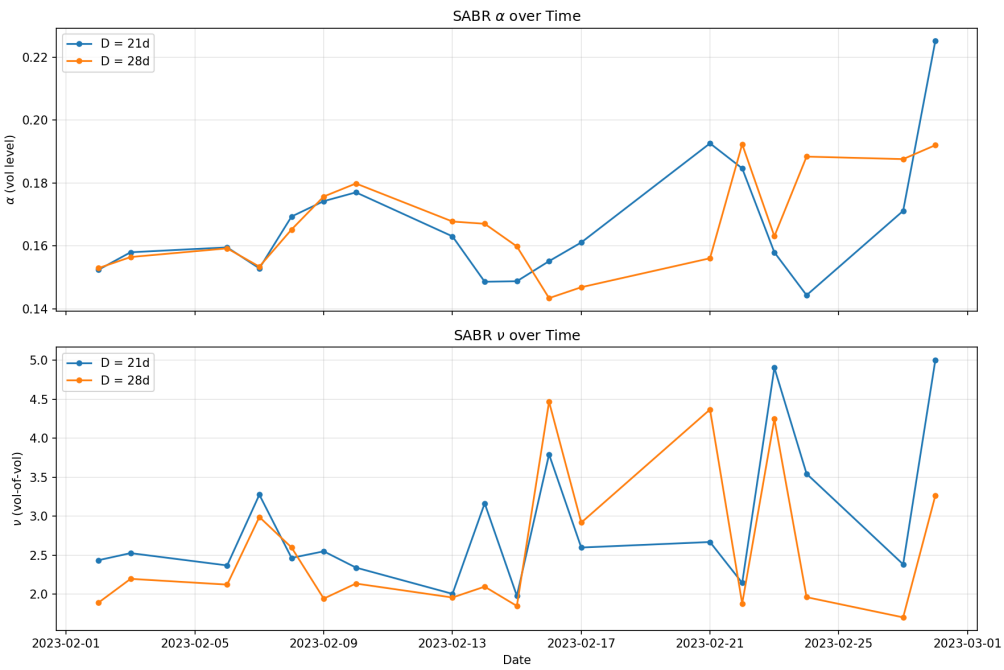

Figure 12: Time series of calibrated SABR parameters $\hat{\alpha}_{t,D}$ (top) and $\hat{\nu}_{t,D}$ (bottom) for maturities $D = 21$ days (blue) and $D = 28$ days (orange) over February 2023.

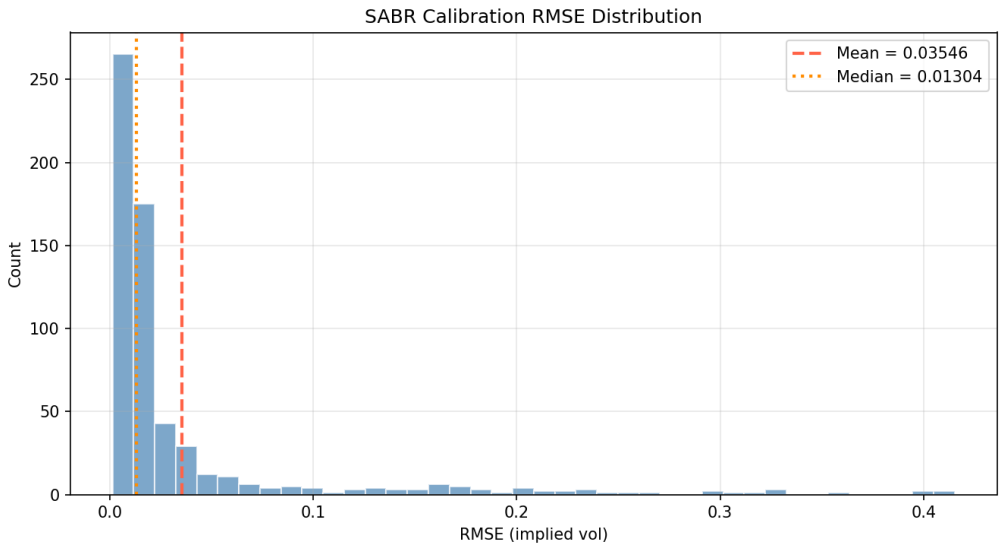

Figure 13: Distribution of SABR calibration RMSE across all 609 slices. Median RMSE = 0.013; mean RMSE = 0.035.

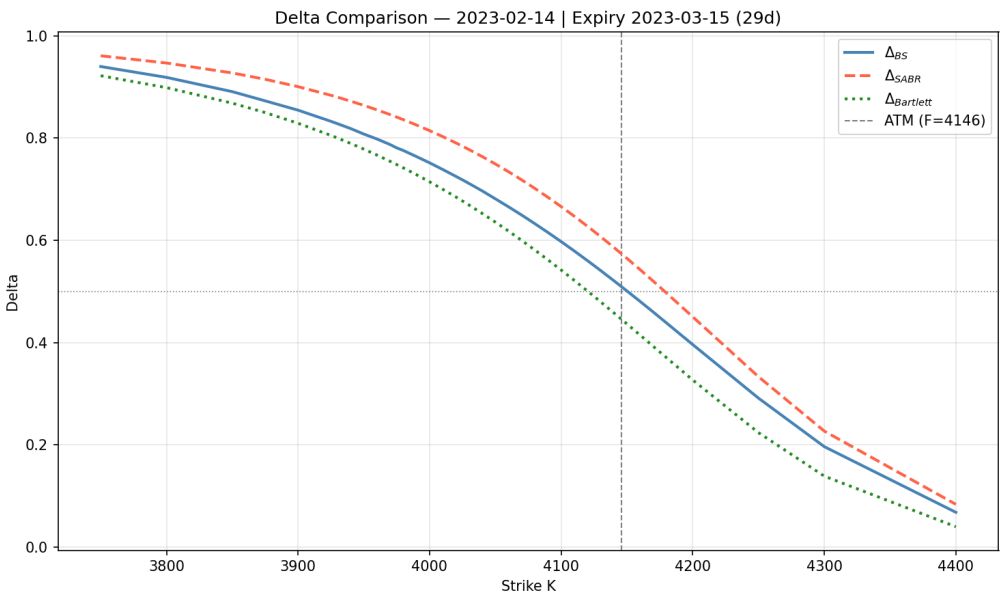

Figure 14: $\Delta^{\mathrm{BS}}(K)$ (blue solid), $\Delta^{\mathrm{SABR}}(K)$ (red dashed), and $\Delta^{\mathrm{Bart}}(K)$ (green dotted) as a function of strike K. Date: 14 February 2023, expiry 15 March 2023 ($D = 29$ days, $F = 4{,}146$, $\hat{\rho} = -0.530$).

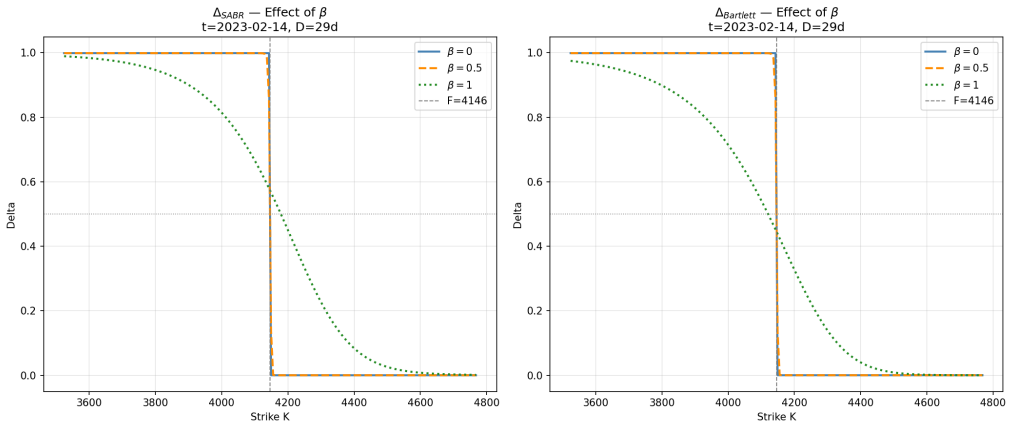

Figure 15: $\Delta^{\mathrm{SABR}}(K)$ (left) and $\Delta^{\mathrm{Bart}}(K)$ (right) for $\beta \in \{0, 0.5, 1\}$. Date: 14 February 2023, $D = 29$ days. Parameters fixed at values calibrated for $\beta = 1$.

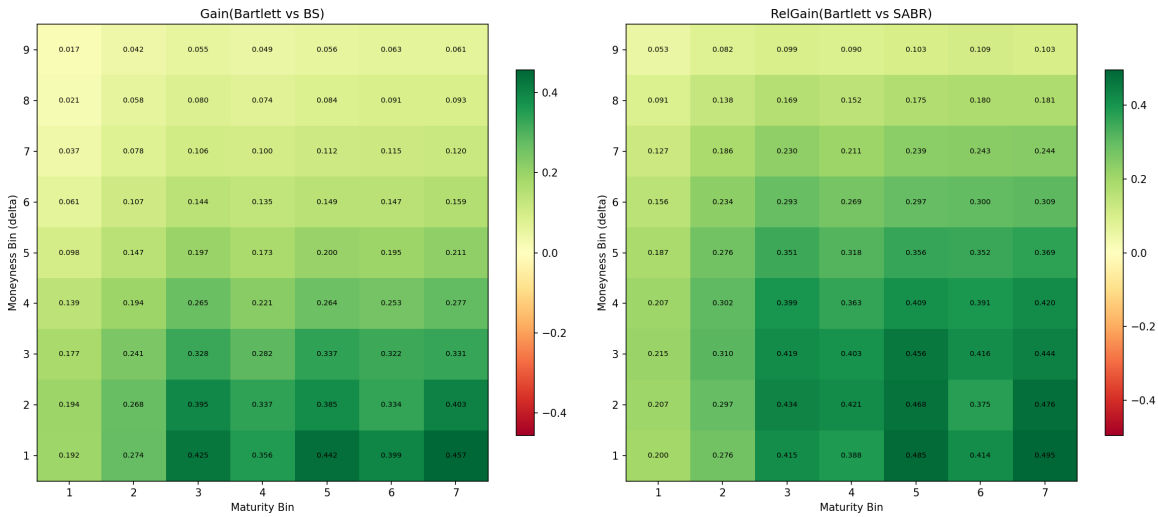

Figure 16: Bucketed Gain of Bartlett vs BS (left) and relative Gain of Bartlett vs SABR (right) across the 9×7 moneyness–maturity grid. Moneyness bins based on $\Delta^{\mathrm{BS}} \in (0.05, 0.95)$ with width 0.10; maturity bins on $D \in (14, 200)$ days with width ≈ 26.6 days.